

SOFIANE MALLEM

Software Developer

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SUMMARY

Software developer with one year of professional experience, skilled in Python automation, backend development, and data pipeline engineering. Focused on building tools that improve workflows and increase operational efficiency.

EXPERIENCE

Synthavo
Technical Customer Success

Nov 2025 - Apr 2026

- Prepared, cleaned, and transformed customer CAD data types for use in internal data-gen pipelines, ensuring onboarding process stays under 2-3 days for new data.
- Onboarded new customers and new customer data, ensuring data quality and compatibility across 10+ servers and dozens of internal service workers.
- Developed, automated and maintained Python tooling to improve pipeline reliability, scalability, and ensure process efficiency.
- Monitored and managed distributed rendering infrastructure, proactively addressing bottlenecks and maintaining >99% service uptime.
- Served as a technical point of contact for complex issues, ensuring fast and smooth communication and minimal service disruption.

Synthavo
Technical Customer Success Intern

Feb 2025 - Aug 2025

- Developed internal Python scripts and automated processes to support CAD data processing and internal data-gen workflows.
- Migrated and optimized Blender-based rendering services, resulting in approximately 40% faster rendering times and improved performance.
- Built and maintained workflows for cleaning and transforming customer CAD data, ensuring high-quality input for production pipelines.
- Streamlined the onboarding process for new customer data, reducing the timeline from ~5 to 2-3 days through careful planning, improved tooling, and automated process enhancements.
- Provided ongoing support for production systems by diagnosing and resolving pipeline and infrastructure issues, maintaining stability and continuity of service.

PROJECTS

Hang — Distributed Real-Time Messaging Platform

FASTAPI, WEBSOCKETS, REDIS PUB/SUB, POSTGRESQL, DOCKER, NGINX

- Designed and implemented a real-time messaging platform supporting conversation-based communication, typing indicators, read receipts, and message history.
- Built a distributed WebSocket architecture using Redis Pub/Sub to synchronize events across multiple FastAPI instances.
- Implemented horizontal scaling behind an Nginx load balancer with stateless application nodes deployed through Docker containers.
- Developed cursor-based message pagination with scroll-triggered loading to efficiently handle large conversation histories.
- Integrated Google OAuth authentication and S3-compatible object storage for user identity management and media handling.
- Created automated Docker-based local infrastructure including PostgreSQL, Redis, Prometheus, multi-instance application deployment, and startup orchestration scripts.

EDUCATION

42 Heilbronn
Software Engineering Course

Feb 2023 - Dec 2025
Heilbronn, Germany

University of S.T Houari Boumedien
Bachelor of Computer Science

Sep 2017 - Jun 2020
Algiers, Algeria

LANGUAGE

English (Native) French (Native) Arabic (Native)

REFERENCES

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